Spatial Coding Summary

JPEG

An Image Compression Standard

JPEG: A Foundation of Image Compression

- **JPEG:** **J**oint **P**hotographic **E**xperts **G**roup — the committee that developed the JPEG image compression standard.
  - JPEG is a lossy image compression standard widely used for digital images.
  - It exploits: 1. human visual perception characteristics
    2. spatial redundancy in images
  - JPEG compresses each image independently.
  - Modern video codecs extend this idea by additionally exploiting temporal redundancy.

JPEG

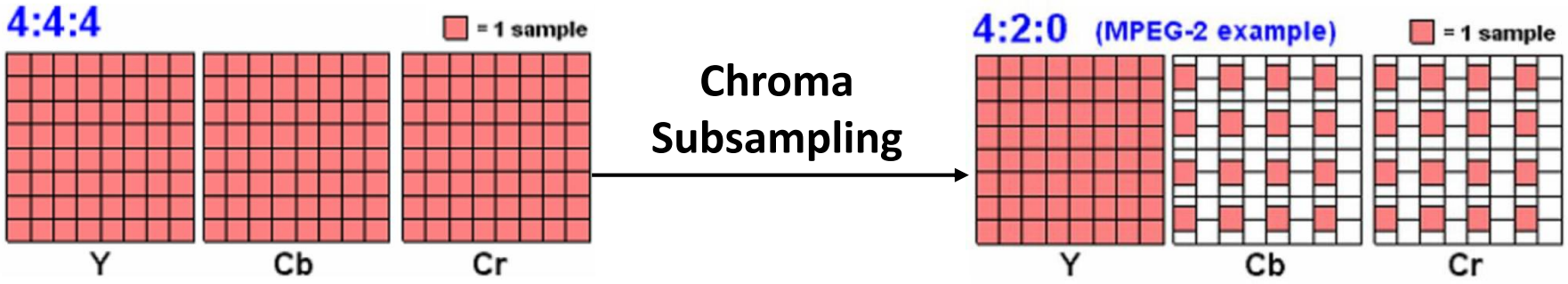
- **Encoding Pipeline**
  1. **Color Space Conversion: RGB to YCbCr** (no loss, no data reduced)
  2. **Color Sub-sampling (**information lost, data reduced**)**
  3. **Discrete Cosine Transform (DCT)** (no loss, no data reduced)
  4. **Quantization** (information lost, data reduced)
  5. **ZigZag Scaning** (no loss, no data reduced)
  6. **Entropy Coding** (no loss, data reduced)
     - **DPCM** (Differential Pulse Code Modulation)
     - **RLC** (Run Length Coding)
     - **VLC** (Variable Length Coding)

JPEG

- **Color Space Conversion**
  - Convert the image from RGB to YCbCr ➔ RGB → YCbCr
    - Y: luminance (brightness)
    - CbCr: chrominance (color information)
  - RGB components are highly correlated.
  - Transforming to YCbCr separates luminance (brightness) from chrominance (color).
  - Human vision is more sensitive to brightness changes than color changes. Separating components according to perceptual importance makes compression more efficient.
  - The color-space conversion itself is essentially reversible: RGB ↔ YCbCr
    - Only negligible errors may occur due to rounding and finite-precision arithmetic.

JPEG

- **Color Sub-sampling**
  - Use low resolution for **chrominance** because human vision is more sensitive to brightness changes than color changes.
  - **4:2:0:** Cb and Cr are sampled at one quarter resolution of Y
    - Each 8x8 DCT block → 4Y + 1Cr + 1Cb



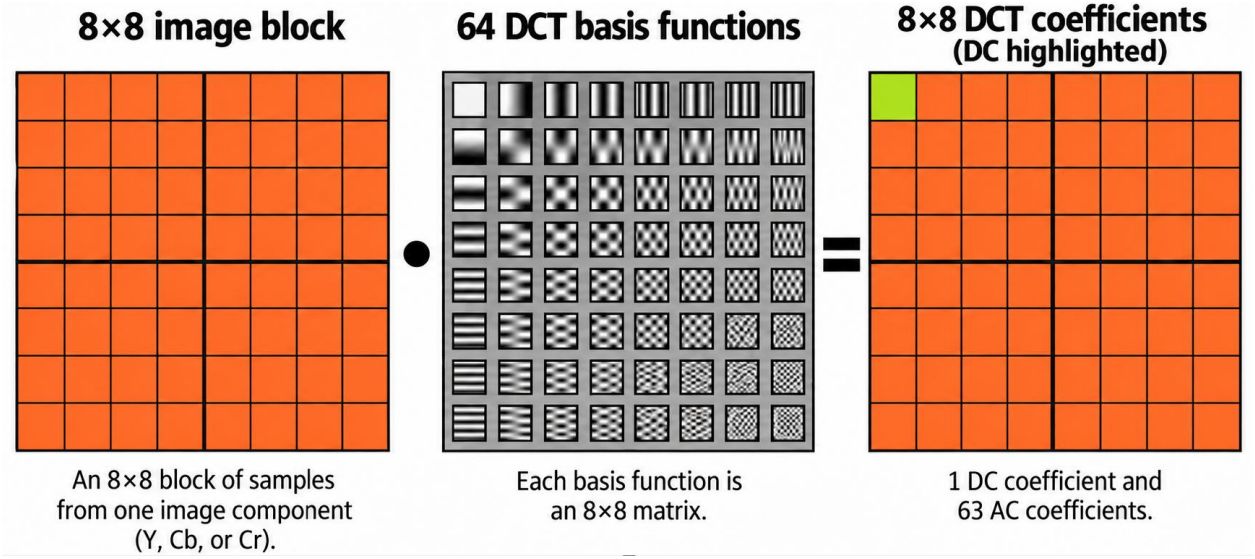
- **Effect**
  - Less data to encode
  - Small visual impact

JPEG

- **DCT (Discrete cosine transform)**

$$F(u,v) = \frac{C(u)C(v)}{4}\sum_{i=0}^{7}\sum_{j=0}^{7}\cos\frac{(2i+1)u\pi}{16}\cos\frac{(2j+1)v\pi}{16}f(i,j) \qquad C(\xi) = \begin{cases} \frac{\sqrt{2}}{2} & \text{if } \xi = 0 \\ 1 & \text{otherwise} \end{cases}$$

  - JPEG applies a 2D DCT to each 8x8 block.
  - The transform uses 64 basis functions, each basis is an 8x8 matrix.
  - An 8x8 block of pixels is transformed into 64 coefficients.
  - The 2D DCT is separable: apply 1D DCT to all rows, then apply 1D DCT to all columns

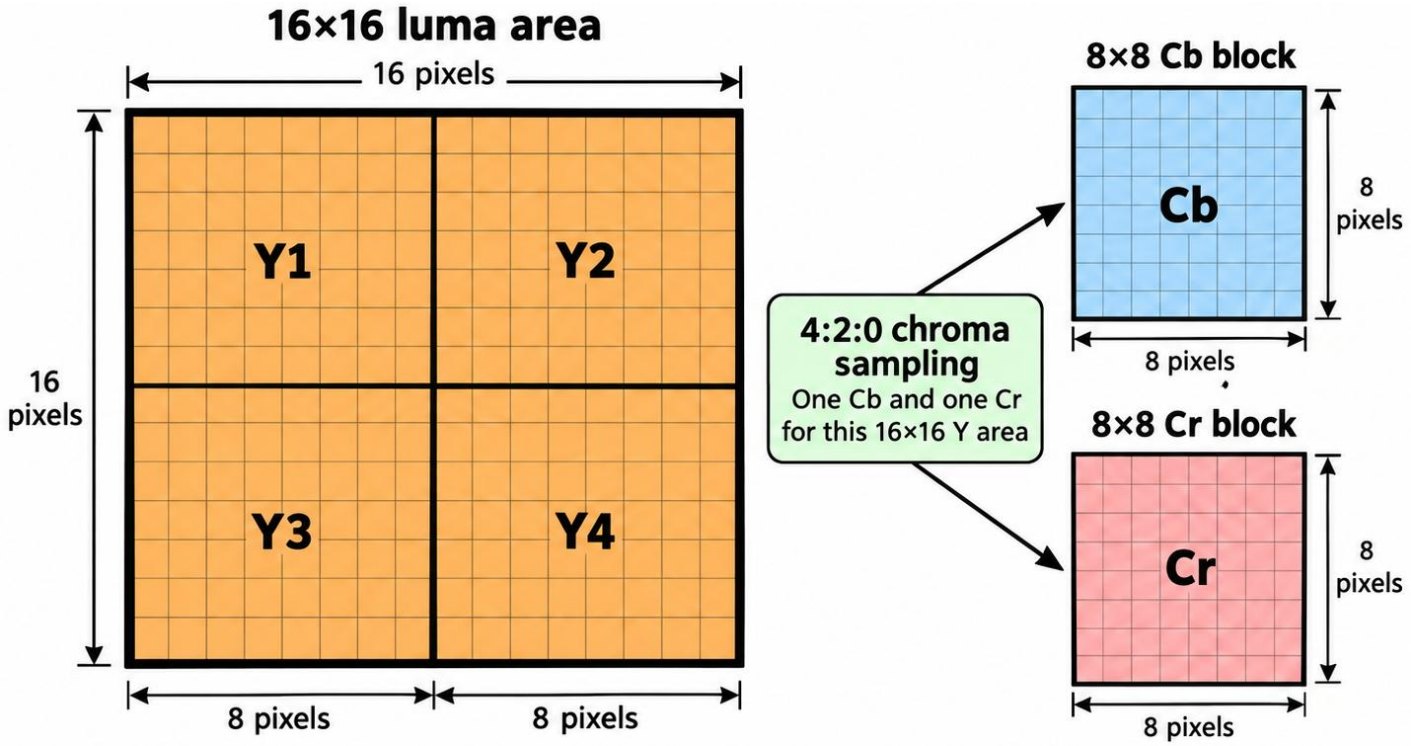


For a color image using 4:2:0 chroma subsampling, how are the Y, Cb, and Cr blocks organized for DCT?

JPEG

- **MCU (Minimum Coded Unit)**
  - For JPEG 4:2:0, one 16x16 MCU contains 4 Y blocks, 1 Cb block, and 1 Cr block

    → Each block is an 8x8 block and is transformed independently using the 2D DCT.



JPEG

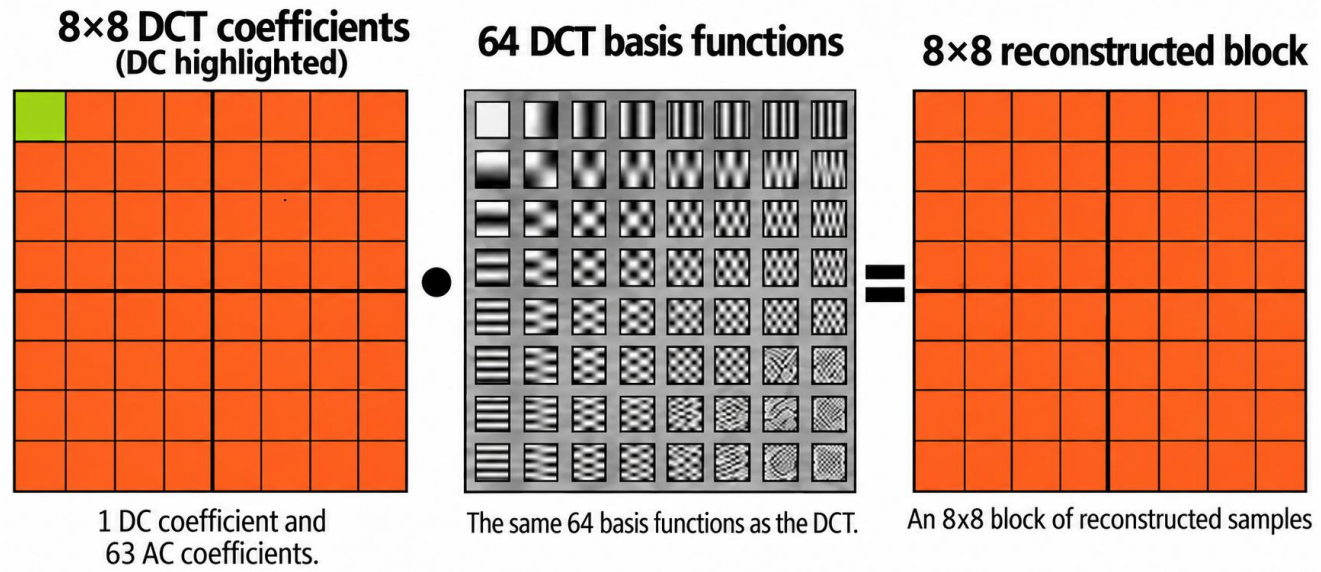
- **IDCT (Inverse Discrete cosine transform)**

IDCT:

$$\tilde{f}(i,j)=\sum_{u=0}^{7}\sum_{v=0}^{7}\frac{C(u)C(v)}{4}\cos\frac{(2i+1)u\pi}{16}\cos\frac{(2j+1)v\pi}{16}F(u,v)$$

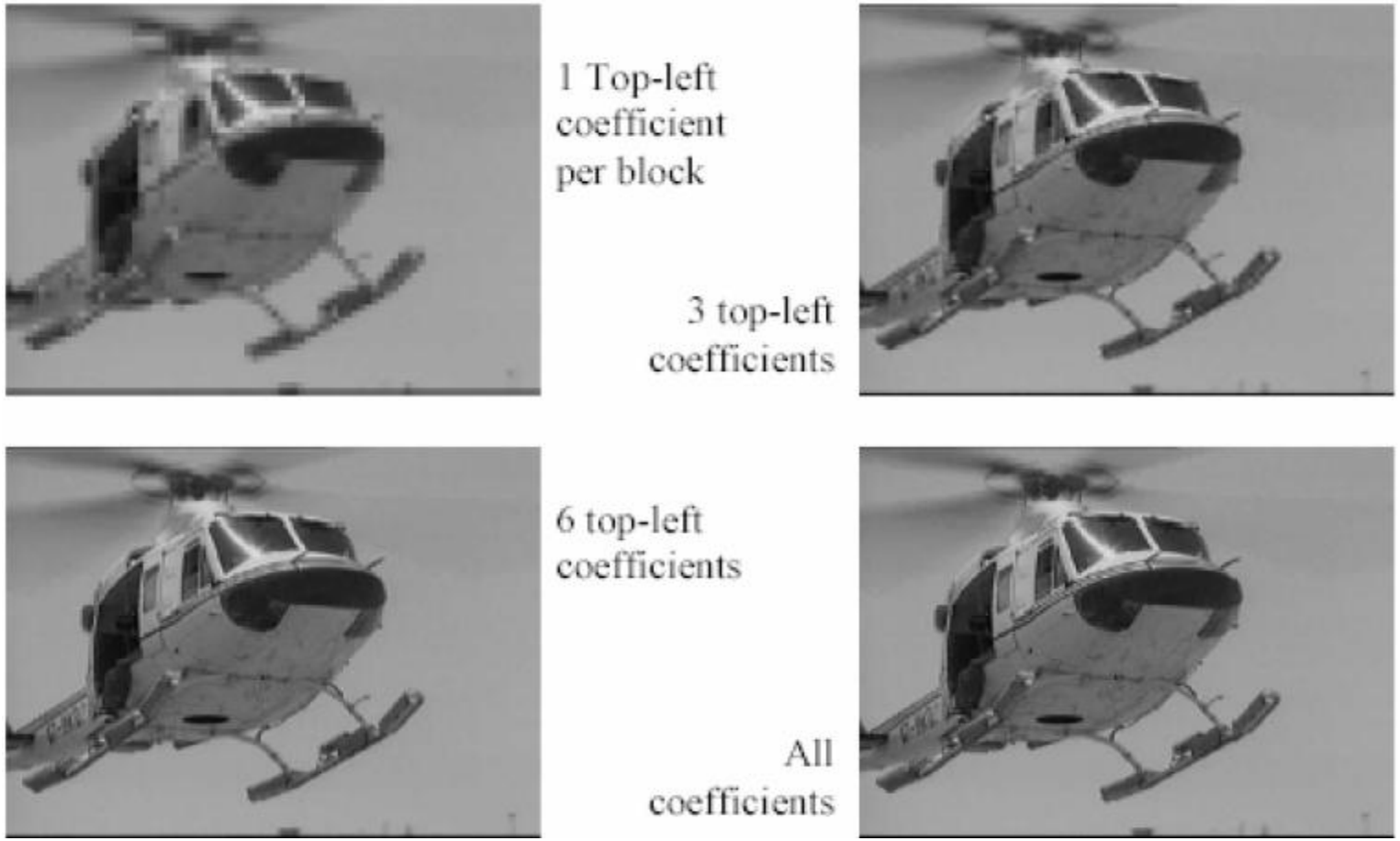
$$C(\xi)=\begin{cases}\frac{\sqrt{2}}{2} & \text{if } \xi=0\\ 1 & \text{otherwise}\end{cases}$$

- Used by the JPEG decoder to reconstruct image blocks.
- Uses the same 64 basis functions as the DCT.
- DCT coefficients are used as weights for the basis functions.
- The weighted basis functions are summed to reconstruct the 8x8 block



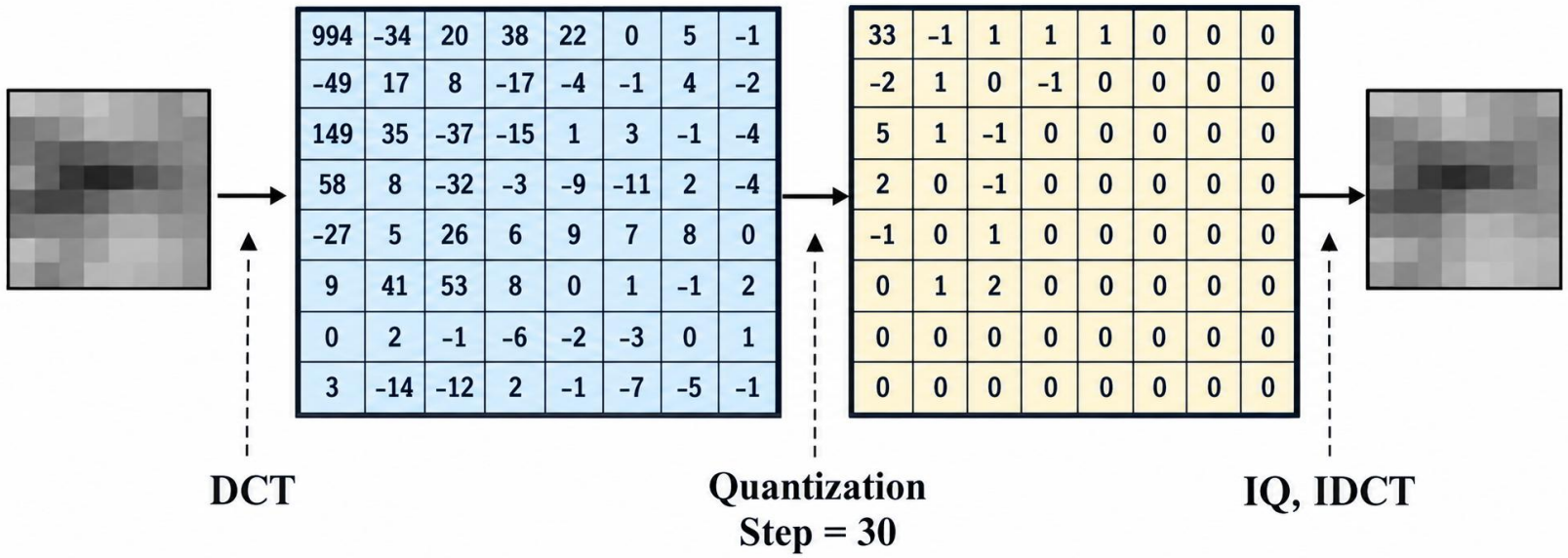
JPEG

- **DCT energy compaction**
  - Most image blocks only contain a few significant coeff. (usually the “lowest frequency coeff”)



JPEG

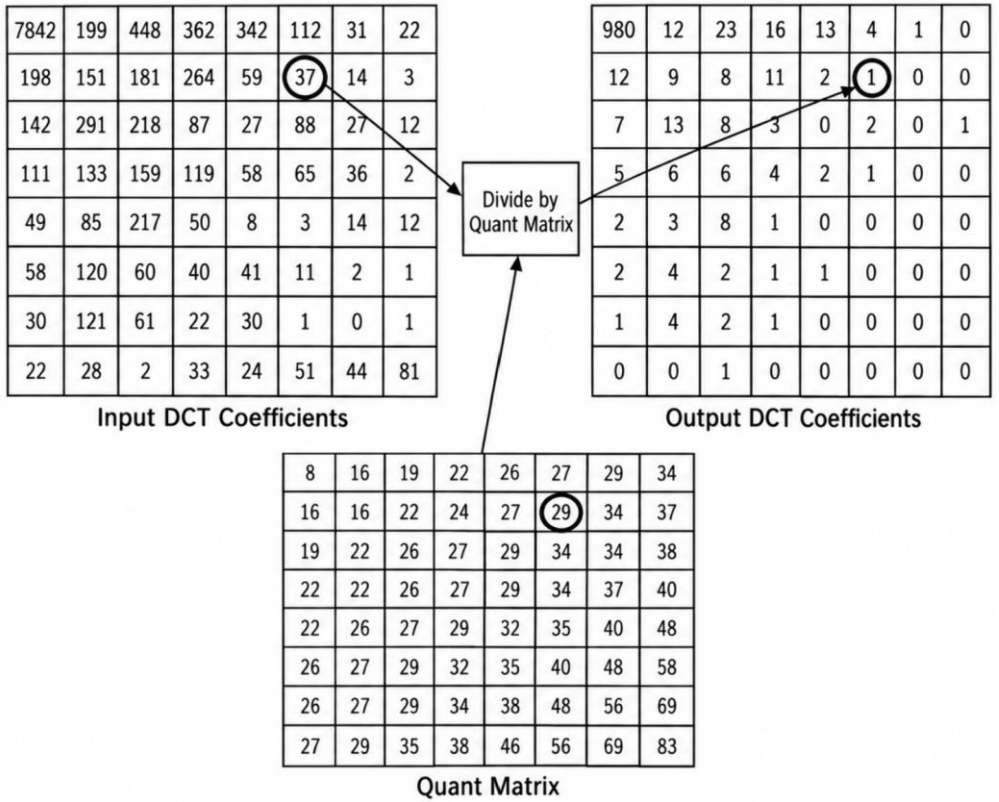
- **Quantization**
  - Represent DCT coefficients with lower precision to reduce the number of bits.
  - Divide each DCT coefficient by a quantization step size and round the result.
  - Typically, many high-frequency coefficients become zero after quantization.
- **Inverse Quantization**
  - Multiply each quantized coefficient by the same step size to approximately recover the original DCT coefficient. (Note: The lost precision cannot be recovered)

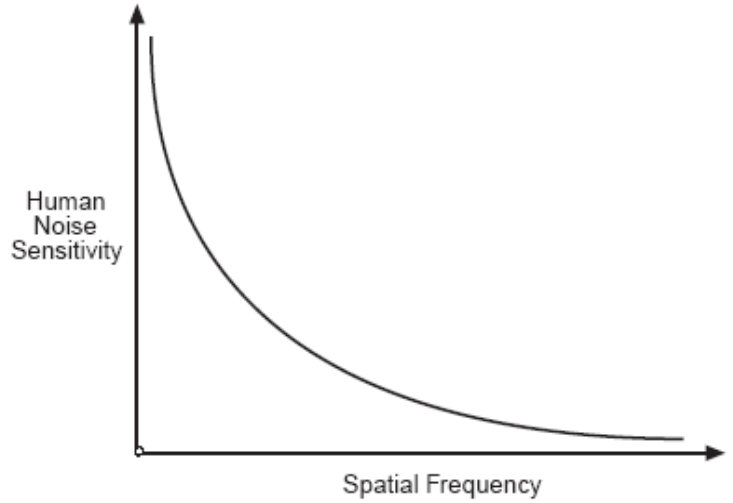


JPEG

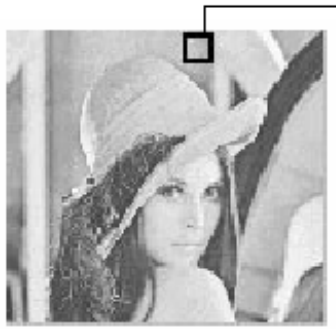
- **Weighted Quantization**
  - Human visual sensitivity varies with spatial frequency.
  - More noise can be tolerated at high frequencies
  - Low-frequency coefficients → small quantization steps
  - High-frequency coefficients → large quantization steps
  - A quantization matrix Q[u,v] defines the step size for each DCT coefficient F[u,v]
    - **Quantized DCT coefficient :**

$$\hat{F}(u,v) = \text{round}\left(\frac{F(u,v)}{Q(u,v)}\right)$$





Example 1: DCT, Quantization, Dequantization, IDCT



Smooth block

200	202	189	188	189	175	175	175
200	203	198	188	189	182	178	175
203	200	200	195	200	187	185	175
200	200	200	200	197	187	187	187
200	205	200	200	195	188	187	175
200	200	200	200	200	190	187	175
205	200	199	200	191	187	187	175
210	200	200	200	188	185	187	186

$f(i,j)$

DCT

515	65	-12	4	1	2	-8	5
-16	3	2	0	0	-11	-2	3
-12	6	11	-1	3	0	1	-2
-8	3	-4	2	-2	-3	-5	-2
0	-2	7	-5	4	0	-1	-4
0	-3	-1	0	4	1	-1	0
3	-2	-3	3	3	-1	-1	3
-2	5	-2	4	-2	2	-3	0

$F(u,v)$

Quantization

$$\hat{F}(u,v) = round\left(\frac{F(u,v)}{Q(u,v)}\right)$$

32	6	-1	0	0	0	0	0
-1	0	0	0	0	0	0	0
-1	0	1	0	0	0	0	0
-1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

$\hat{F}(u,v)$

Dequantization

$$\tilde{F}(u,v) = \hat{F}(u,v)\,Q(u,v)$$

512	66	-10	0	0	0	0	0
-12	0	0	0	0	0	0	0
-14	0	16	0	0	0	0	0
-14	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

$\tilde{F}(u,v)$

IDCT

199	196	191	186	182	178	177	176
201	199	196	192	188	183	180	178
203	203	202	200	195	189	183	180
202	203	204	203	198	191	183	179
200	201	202	201	196	189	182	177
200	200	199	197	192	186	181	177
204	202	199	195	190	186	183	181
207	204	200	194	190	187	185	184

$\tilde{f}(i,j)$

1	6	-2	2	7	-3	-2	-1
-1	4	2	-4	1	-1	-2	-3
0	-3	-2	-5	5	-2	2	-5
-2	-3	-4	-3	-1	-4	4	8
0	4	-2	-1	-1	-1	5	-2
0	0	1	3	8	4	6	-2
1	-2	0	5	1	1	4	-6
3	-4	0	6	-2	-2	2	2

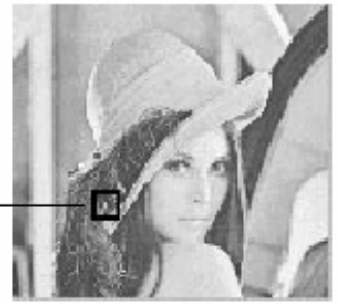
Error

$$\epsilon(i,j) = f(i,j) - \tilde{f}(i,j)$$

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	35	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

$Q(u,v)$

Example 2: DCT, Quantization, Dequantization, IDCT



Textured block

$$f(i,j)$$

70	70	100	70	87	87	150	187
85	100	96	79	87	154	87	113
100	85	116	79	70	87	86	196
136	69	87	200	79	71	117	96
161	70	87	200	103	71	96	113
161	123	147	133	113	113	85	161
146	147	175	100	103	103	163	187
156	146	189	70	113	161	163	197

DCT

-80	-40	89	-73	44	32	53	-3
-135	-59	-26	6	14	-3	-13	-28
47	-76	66	-3	-108	-78	33	59
-2	10	-18	0	33	11	-21	1
-1	-9	-22	8	32	65	-36	-1
5	-20	28	-46	3	24	-30	24
6	-20	37	-28	12	-35	33	17
-5	-23	33	-30	17	-5	-4	20

$$F(u,v)$$

Error

0	10	-6	-24	25	-16	4	11
0	-1	11	4	-15	27	-6	-31
2	-14	24	-23	-4	-11	-3	29
4	16	-24	20	24	1	11	-49
-12	13	-27	-7	-8	-18	12	23
-3	0	16	-2	-20	21	0	-1
5	-12	6	27	-3	2	14	-37
6	5	-6	-9	6	14	-47	44

$$\epsilon(i,j) = f(i,j) - \tilde{f}(i,j)$$

16	11	10	16	24	40	51	61
12	12	14	19	26	58	60	55
14	13	16	24	40	57	69	56
14	17	22	29	51	87	80	62
18	22	37	56	68	109	103	77
24	35	55	64	81	104	113	92
49	64	78	87	103	121	120	101
72	92	95	98	112	100	103	99

$$Q(u,v)$$

$$\hat{F}(u,v) = round\left(\frac{F(u,v)}{Q(u,v)}\right)$$

Quantization

-5	-4	9	-5	2	1	1	0
-11	-5	-2	0	1	0	0	-1
3	-6	4	0	-3	-1	0	1
0	1	-1	0	1	0	0	0
0	0	-1	0	0	1	0	0
0	-1	1	-1	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

$$\hat{F}(u,v)$$

$$\tilde{F}(u,v) = \hat{F}(u,v)\,Q(u,v)$$

Dequantization

-80	-44	90	-80	48	40	51	0
-132	-60	-28	0	26	0	0	-55
42	-78	64	0	-120	-57	0	56
0	17	-22	0	51	0	0	0
0	0	-37	0	0	109	0	0
0	-35	55	-64	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

$$\tilde{F}(u,v)$$

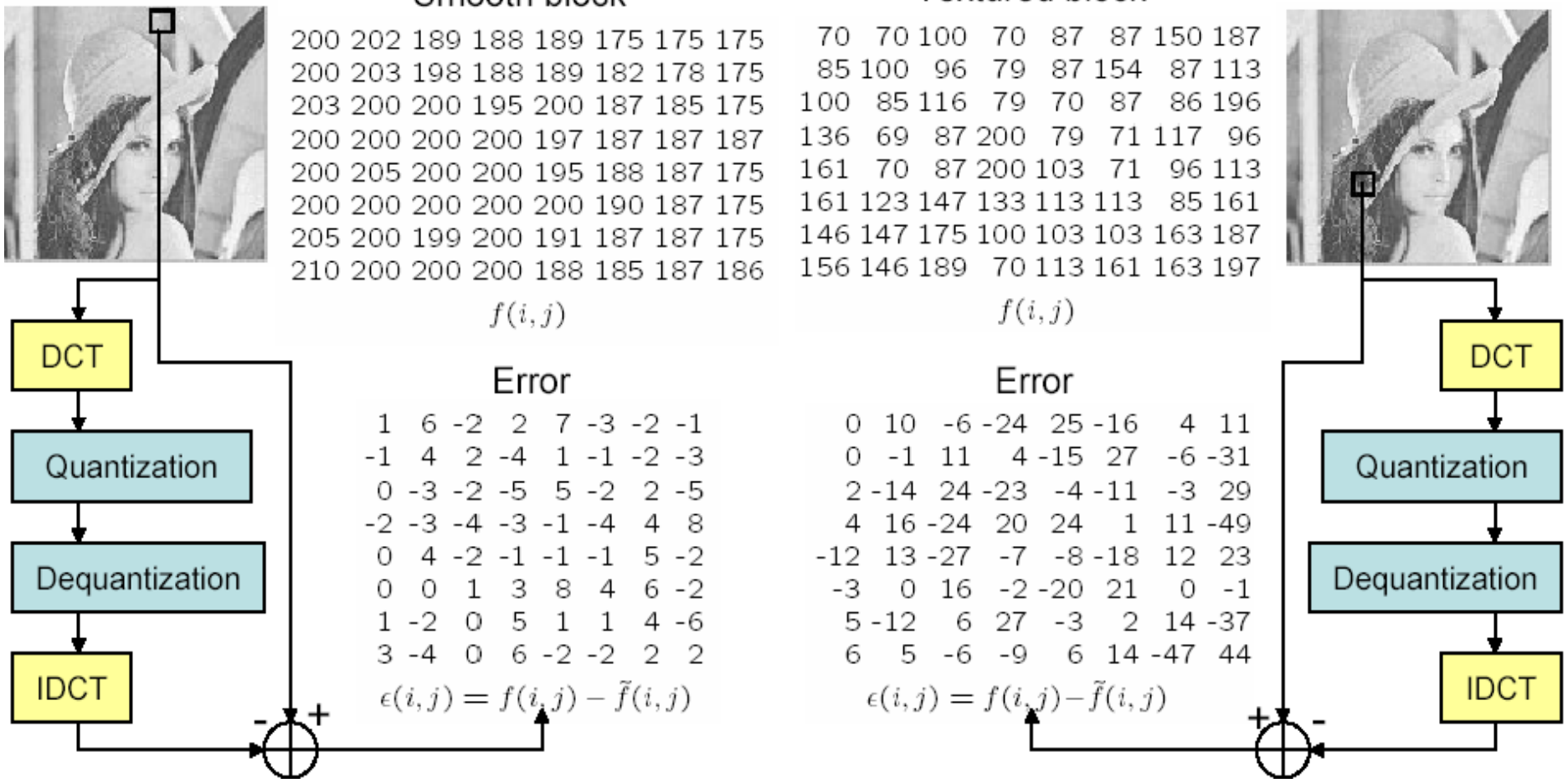
IDCT

70	60	106	94	62	103	146	176
85	101	85	75	102	127	93	144
98	99	92	102	74	98	89	167
132	53	111	180	55	70	106	145
173	57	114	207	111	89	84	90
164	123	131	135	133	92	85	162
141	159	169	73	106	101	149	224
150	141	195	79	107	147	210	153

$$\tilde{f}(i,j)$$

Comparison of two examples

- **The error is larger for the textured block so more loss is introduced.**



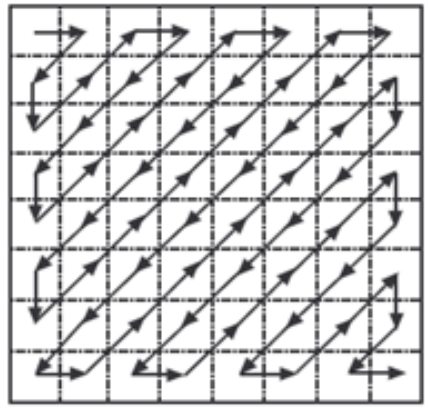
JPEG

- **Zig-zag Scanning**
  - After weighted quantization, low-value coef. might be truncated to zero. More efficient transmission can be obtained if all the non-zero coef. are sent first, followed by a code indicating that the remainder are all zero. Zig-zag scanning increases the probability to achieve this.

Quantized DCT coefficients

32	6	-1	0	0	0	0	0
-1	0	0	0	0	0	0	0
-1	0	1	0	0	0	0	0
-1	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0

Zigzag



32 6 -1 -1 0 -1 0 0 0 -1 0 0 1 0 0 0 • • •

JPEG

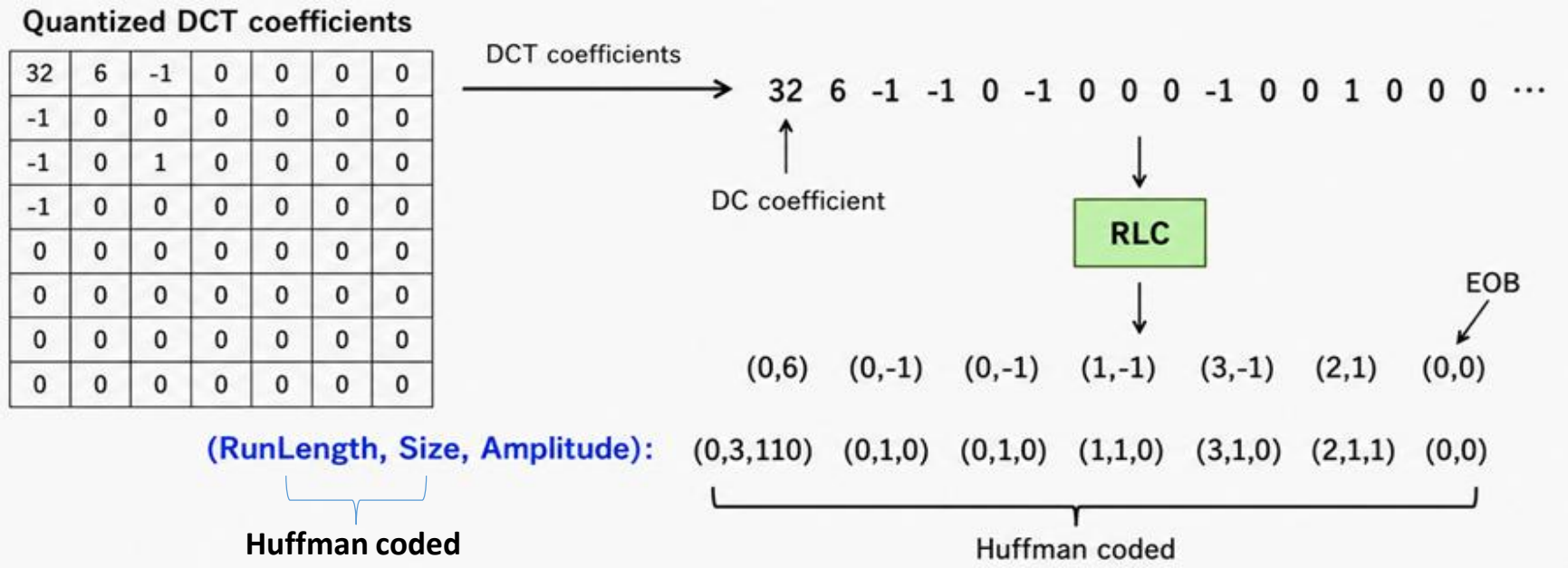
- **Entropy Coding**

  After zig-zag scanning, many AC coefficients become zero, but non-zero values may still appear between them. To encode these coefficients efficiently, MPEG-2 uses **DPCM, RLC, and VLC**.

  - **RLC** (Run Length Coding)
    - Encodes consecutive zero **AC coefficients** by recording the number of zeros before the next non-zero value.
  - **DPCM** (Differential Pulse Code Modulation)
    - Encodes **DC coefficients** by transmitting the difference between neighboring blocks, since DC values usually change slowly.
  - **VLC** (Variable Length Coding)
    - Assigns short codewords to frequent symbols and long codewords to rare symbols for further compression..

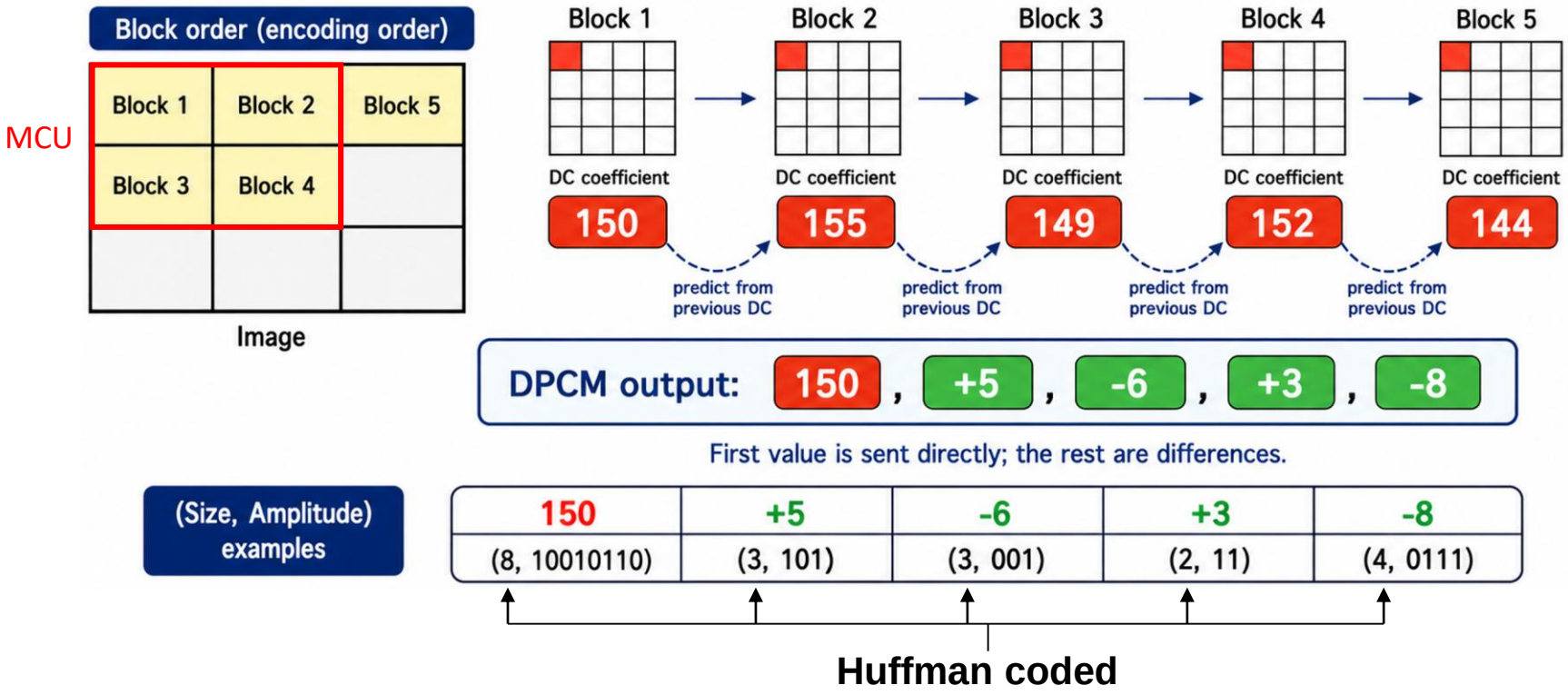
JPEG

- **RLC (Run Length Coding)**
  - Used to code AC coefficients for each block
  - (RunLength,Value) : RunLength is the number of zeros to skip, Value is the next non-zero value.
  - (RunLength, Size, Amplitude) : Size is the number of bits needed for representing Amplitude, and the Amplitude is the actual bits.



JPEG

- **DPCM (Differential Pulse Code Modulation)**
  - Used to encode DC coefficients for each block
  - Instead of coding the absolute DC value, the encoder codes the difference from the previous block's DC coefficient.

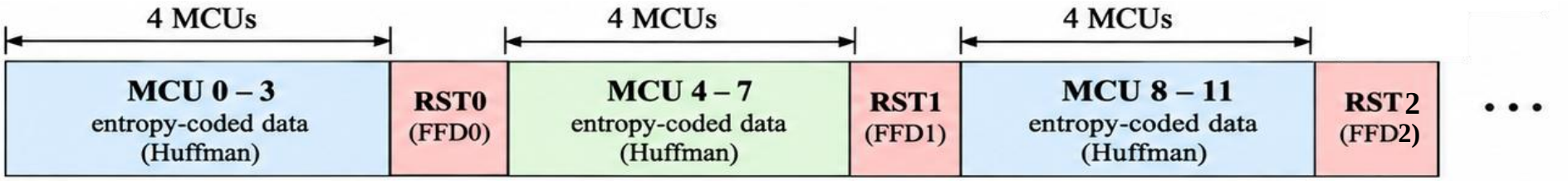


JPEG

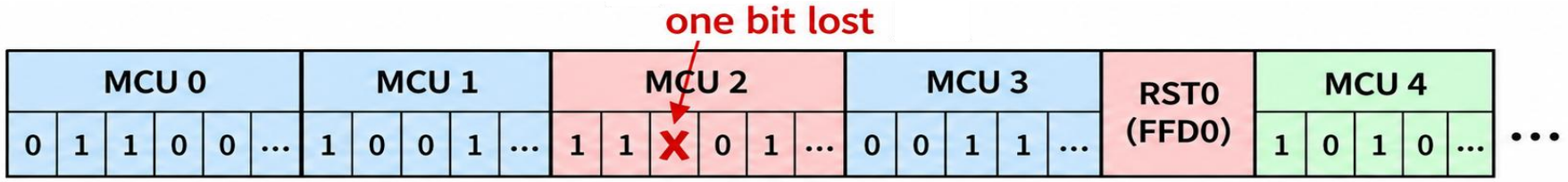
- **How JPEG Limits Error Propagation in DPCM**
  - JPEG insert **restart markers** periodically.
  - A **restart interval** specifies how many MCUs are coded between restart markers.
  - At each restart marker:
    - DC predictors are reset.
    - Entropy decoding is resynchronized.
    - Error propagation is limited to the current restart interval.
  - JPEG defines 8 restart markers:
    - RST0 → RST1 → RST2 → … → RST7 → RST0 (cyclically used)
    - The encoder chooses a **restart interval N,** specified by the DRI marker.
    - A restart marker is inserted every N MCUs.
    - JPEG uses special restart marker codes (RST0-RST7) to distinguish them from entropy-coded data.

Example with Restart Marker

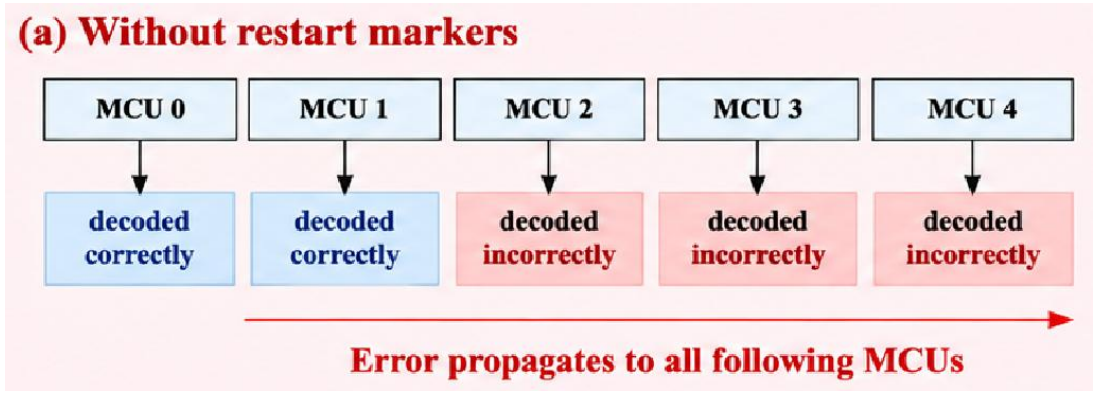
1. **JPEG bitstream with restart markers** (restart interval = 4 MCUs)

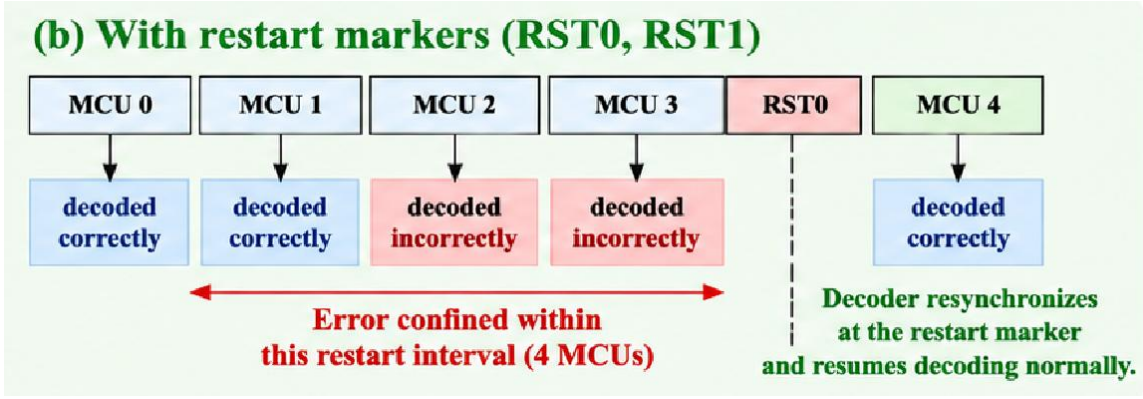


2. **Bit error in the stream** (one bit is lost in MCU2)



3. **Decoding**





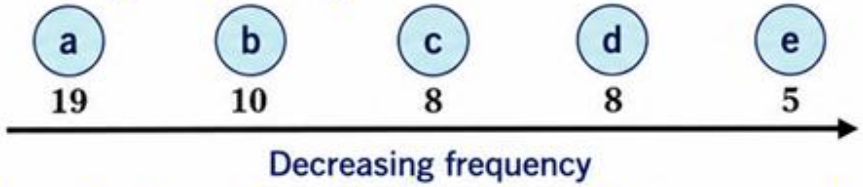
JPEG

- **VLC (Variable Length Coding)**
  - **Huffman Coding**

1. Show symbols as leaf nodes

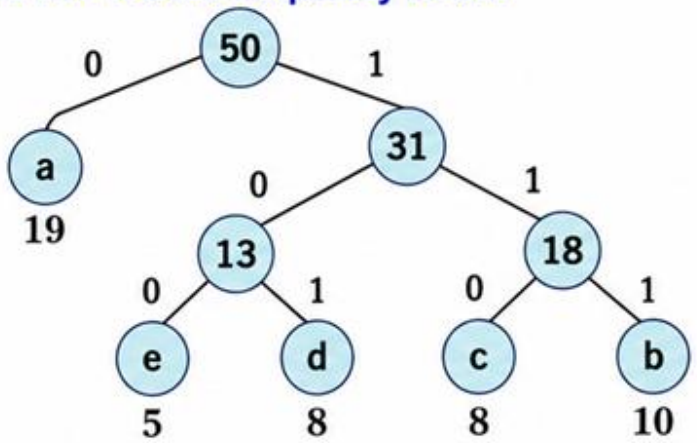


2. Sort by frequency



3. Repeatedly combine the two lowest-frequency nodes

- Combine e(5) and d(8) → node 13
- Combine c(8) and b(10) → node 18
- Combine 13 and 18 → node 31
- Combine a(19) and 31 → root 50



Symbols	a	b	c	d	e
Frequency	19	10	8	8	5

More frequent symbols are placed closer to the root, so they use fewer bits.

Symbol	Code
a	0
b	111
c	110
d	101
e	100

JPEG Coding Pipeline

